Final Project

Analysis of Childhood Homelessness by Children Who Have Lived with Someone with a Drug/Alcohol Problem

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Setup and Data Wrangling:

Setup, importing the data, and selecting the necessary columns for our tests:

```
# Install relevant packages
if (!require("pacman")) install.packages("pacman")
```

```
Loading required package: pacman
```

```
pacman::p_load(
  marginaleffects, tidyverse, haven, knitr, gtsummary,
  Hmisc, janitor, broom, pROC, labelled, lmtest
)

# Load the dataset
nsch2023data <- readRDS("nsch2023data.rds") %>%

# Select the necessary columns
select("ACE9", "EVERHOMELESS", "SC_RACE_R", "SC_AGE_YEARS", "SC_SEX") %>%
  rename(Race = "SC_RACE_R",
         Drug_Alc_Prob = "ACE9",
         Age = "SC_AGE_YEARS",
         Sex = "SC_SEX",
         EverHomeless = "EVERHOMELESS")
```

Data Wrangling: Renaming variables to be more clear, converting categorical variables to factor variables, and releveling `Drug_Alc_Prob` and `EverHomeless` to set the referent groups to the ones that have not lived with someone with a drug/alcohol problem and have not experienced homelessness.

```
dat <- nsch2023data %>%
mutate(
Race = factor(Race, levels = c(1,2,3,4,5,7),
                       labels = c("White",
                                  "Black",
                                  "Native American",
                                  "Asian",
                                  "Pacific Islander",
                                  "Two or More Races")),
Drug_Alc_Prob = relevel(factor(Drug_Alc_Prob,
                 levels = c(1,2),
                 labels = c("Has", "Has Not")),
          ref = "Has Not"),
EverHomeless = relevel(factor(EverHomeless, levels = c(1,2,3),
                       labels = c("Yes", "No", "Unknown")),
              ref = "No"),
Sex = factor(Sex, levels = c(1,2),
                       labels = c("Male", "Female"))
)
```

Variable	**N**	**N = 55,162**[1]
EverHomeless	53,699	
No		52,572 (98%)
Yes		889 (1.7%)
Unknown		238 (0.4%)
Missing Data		1,463
Drug_Alc_Prob	52,952	
Has Not		48,617 (92%)
Has		4,335 (8.2%)
Missing Data		2,210
Sex	55,162	
Male		28,330 (51%)
Female		26,832 (49%)
Age	55,162	
Mean (SD)		8.3 (5.3)
Median (Q1-Q3)		8.0 (4.0-13.0)
Min-Max		0.0-17.0
Race	55,162	
White		42,095 (76%)
Black		3,686 (6.7%)
Native American		507 (0.9%)
Asian		3,632 (6.6%)
Pacific Islander		260 (0.5%)
Two or More Races		4,982 (9.0%)

[1] n (%)

Descriptive Data Visualization:

Creating a Table 1 to visualize our data, look for any potential missing/problematic values, and get a general idea of the data we are working with.

```
tbl_summary(
  dat,
  include = c(EverHomeless, Drug_Alc_Prob, Sex, Age, Race),
  type = all_continuous() ~ "continuous2",
  statistic = all_continuous() ~ c("{mean} ({sd})",
                                   "{median} ({p25}-{p75})",
                                   "{min}-{max}"),
  missing_text = "Missing Data") |>
add_n() |> # add column with total number of non-missing observations
modify_header(label = "**Variable**") |>
modify_caption("**Table 1: Patient Characteristics**")
```

Moving on as there are missing values but they are not large enough to impact our analysis, and there are no problematic values (other than possibly low sample sizes for Native Americans and Pacific Islanders).

Creating a bivariate tabular analysis to get a sense of how our exposure and outcome variable interact. Using row percentages to get a better sense of the proportion of children who have experienced homelessness between different levels of our exposure variable.

```
dat %>%
  tabyl(Drug_Alc_Prob, EverHomeless) %>%
  adorn_totals(where = c("row", "col")) %>%
  adorn_percentages("row") %>%
  adorn_pct_formatting(digits = 1) %>%
  adorn_ns("front") %>%
  kable()
```

Drug_Alc_Prob	No	Yes	Unknown	NA_	Total
Has Not	47,901 (98.5%)	433 (0.9%)	87 (0.2%)	196 (0.4%)	48,617 (100.0%)
Has	3,762 (86.8%)	410 (9.5%)	138 (3.2%)	25 (0.6%)	4,335 (100.0%)
NA	909 (41.1%)	46 (2.1%)	13 (0.6%)	1,242 (56.2%)	2,210 (100.0%)
Total	52,572 (95.3%)	889 (1.6%)	238 (0.4%)	1,463 (2.7%)	55,162 (100.0%)

Creating and Analyzing an Unadjusted Model:

Creating model1, our unadjusted bivariate model. This will show the relationship between these two variables before adjusting for any covariates. This linear regression will be shown in both odds and probabilities for readability's sake.

```
model1 <- (glm(EverHomeless ~ Drug_Alc_Prob, data = dat,
               family = binomial(link = "logit")))
kable(tidy(model1, exponentiate = TRUE, conf.int = TRUE,), digits = 2)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.01	0.04	-102.59	0	0.01	0.01
Drug_Alc_ProbHas	13.42	0.06	40.88	0	11.85	15.20

```
kable(predictions(model1, by = "Drug_Alc_Prob"))
```

Drug_Alc_Prob	estimate	std.error	statistic	p.value	s.value	conf.low	conf.high	df
Has Not	0.0107391	0.0004684	22.92759	0	384.0415	0.0098211	0.0116572	Inf
Has	0.1271462	0.0050744	25.05636	0	457.8525	0.1172005	0.1370918	Inf

This shows that living with a drug/alcohol problem appears to be significantly associated with an increased likelihood of ever having experienced homelessness.

Setup for an Adjusted Model:

Creating Table 2 to further visualize our data before attempting to create an adjusted model. We will analyze this table and (once again) look for any missing values, problematic values, and to begin investigating statistical significance. This is where we will be able to see if the small values for Native Americans and Pacific Islanders will be an issue.

```
tbl_summary(
  dat,
  by = EverHomeless,
  include = c(Drug_Alc_Prob, Sex, Age, Race),
  type = all_continuous() ~ "continuous2",
  statistic = all_continuous() ~ c("{mean} ({sd})", "{median}
                                    ({p25}-{p75})", "{min}-{max}"),
  missing_text = "Missing Data") |>
add_p(test.args = list(simulate.p.value = TRUE)) |> # Throwing an error and recommending simulatec
modify_header(label = "**Variable**") |>
modify_caption("**Table 2: Patient Characteristics by Homelessness**") |>
bold_labels() |>
modify_spanning_header(c("stat_1", "stat_2") ~ "**Ever Been Homeless**")
```

```
1463 missing rows in the "EverHomeless" column have been removed.
The following errors were returned during `modify_spanning_header()`:
x For variable `Race` (`EverHomeless`) and "estimate", "p.value", "conf.low",
  and "conf.high" statistics: FEXACT error 501. The hash table key cannot be
  computed because the largest key is larger than the largest representable
  int. The algorithm cannot proceed. Reduce the workspace, consider using
  'simulate.p.value=TRUE' or another algorithm.
```

Once again, missing data is not prevalent enough to warrant removing it, and there appears to be no problematic values. while sample sizes are small for Native Americans and Pacific Islanders, they don't seem to be small enough to warrant much concern.

Additionally, Asians have a surprisingly small number in the homeless group; however, this situation mirrors the ones seen in the other two aforementioned races, therefore we will not be changing/binning any values.

Variable	**Ever Been Homeless** **No** N = 52,572[1]	**Yes** N = 889[1]	**Unknown** N = 238[1]	**p-value**[2]
Drug_Alc_Prob				<0.001
Has Not	47,901 (93%)	433 (51%)	87 (39%)	
Has	3,762 (7.3%)	410 (49%)	138 (61%)	
Missing Data	909	46	13	
Sex				0.006
Male	26,953 (51%)	504 (57%)	121 (51%)	
Female	25,619 (49%)	385 (43%)	117 (49%)	
Age				<0.001
Mean (SD)	8.3 (5.3)	11.1 (4.6)	10.5 (4.8)	
Median (Q1-Q3)	8.0 (4.0-13.0)	12.0 (7.0-15.0)	11.0 (7.0-15.0)	
Min-Max	0.0-17.0	0.0-17.0	0.0-17.0	
Race				
White	40,416 (77%)	518 (58%)	145 (61%)	
Black	3,304 (6.3%)	172 (19%)	38 (16%)	
Native American	451 (0.9%)	34 (3.8%)	12 (5.0%)	
Asian	3,486 (6.6%)	18 (2.0%)	17 (7.1%)	
Pacific Islander	224 (0.4%)	12 (1.3%)	3 (1.3%)	
Two or More Races	4,691 (8.9%)	135 (15%)	23 (9.7%)	

[1]n (%)

[2]Pearson's Chi-squared test; Kruskal-Wallis rank sum test; NA

Creating an Adjusted Model

Creating model2, our logistic regression model that adjusts for age, sex, and race as possible covariates. This model will be presented in odds (rather than logodds) for interpretation's sake.

```
model2 <- (glm(EverHomeless ~ Drug_Alc_Prob + Age + Sex + Race,
               data = dat, family = binomial(link = "logit")))
kable(tidy(model2, exponentiate = TRUE, conf.int = TRUE,), digits = 2)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.01	0.09	-59.52	0.00	0.00	0.01
Drug_Alc_ProbHas	11.96	0.07	36.77	0.00	10.48	13.65
Age	1.06	0.01	8.70	0.00	1.05	1.07
SexFemale	0.85	0.06	-2.52	0.01	0.75	0.96
RaceBlack	4.73	0.09	17.43	0.00	3.96	5.62
RaceNative American	3.85	0.18	7.64	0.00	2.70	5.39
RaceAsian	0.99	0.18	-0.04	0.97	0.68	1.40
RacePacific Islander	4.02	0.31	4.54	0.00	2.10	7.06
RaceTwo or More Races	2.08	0.10	7.53	0.00	1.71	2.51

This shows that these covariates have statistically significant impacts on the likelihood of homelessness (except for race=Asian). While statistically significant, covariate influences on homelessness are much smaller than our exposure variable.

Age appears to have an extremely small effect size, but (as it is a continuous variable) it leads to a difference in odds of 16.96 between a 17-year-old and a 1-year-old. While the effect sizes vary, all races (excluding Asian) appear to have a higher likelihood of homelessness compared to White individuals.

Finding the Most Parsimonious Model:

Attempting to use Backwards Selection AIC Minimization to find a more parsimonious model. This will be done with the current adjusted model (model 2) as well as a version of this model that excludes all missing values in the dataset to see if this changes the result.

```
no_na <- dat %>%
  drop_na()
model_no_na <- (glm(EverHomeless ~ Drug_Alc_Prob + Age + Sex + Race,
                data = no_na, family = binomial(link = "logit")))

step(model2, direction = "backward")
```

```
Start:  AIC=8649.74
EverHomeless ~ Drug_Alc_Prob + Age + Sex + Race

                Df Deviance    AIC
<none>               8631.7 8649.7
- Sex            1   8638.1 8654.1
- Age            1   8709.4 8725.4
- Race           5   8948.1 8956.1
- Drug_Alc_Prob  1   9845.3 9861.3


Call:  glm(formula = EverHomeless ~ Drug_Alc_Prob + Age + Sex + Race,
    family = binomial(link = "logit"), data = dat)

Coefficients:
          (Intercept)      Drug_Alc_ProbHas                    Age
            -5.273378              2.481524               0.057365
            SexFemale             RaceBlack    RaceNative American
            -0.162963              1.553791               1.348348
            RaceAsian   RacePacific Islander  RaceTwo or More Races
            -0.006418              1.391054               0.731884

Degrees of Freedom: 52730 Total (i.e. Null);  52722 Residual
  (2431 observations deleted due to missingness)
Null Deviance:      10440
Residual Deviance: 8632     AIC: 8650
```

```
step(model_no_na, direction = "backward")
```

```
Start:  AIC=8649.74
EverHomeless ~ Drug_Alc_Prob + Age + Sex + Race

                Df Deviance    AIC
<none>               8631.7 8649.7
- Sex            1   8638.1 8654.1
```

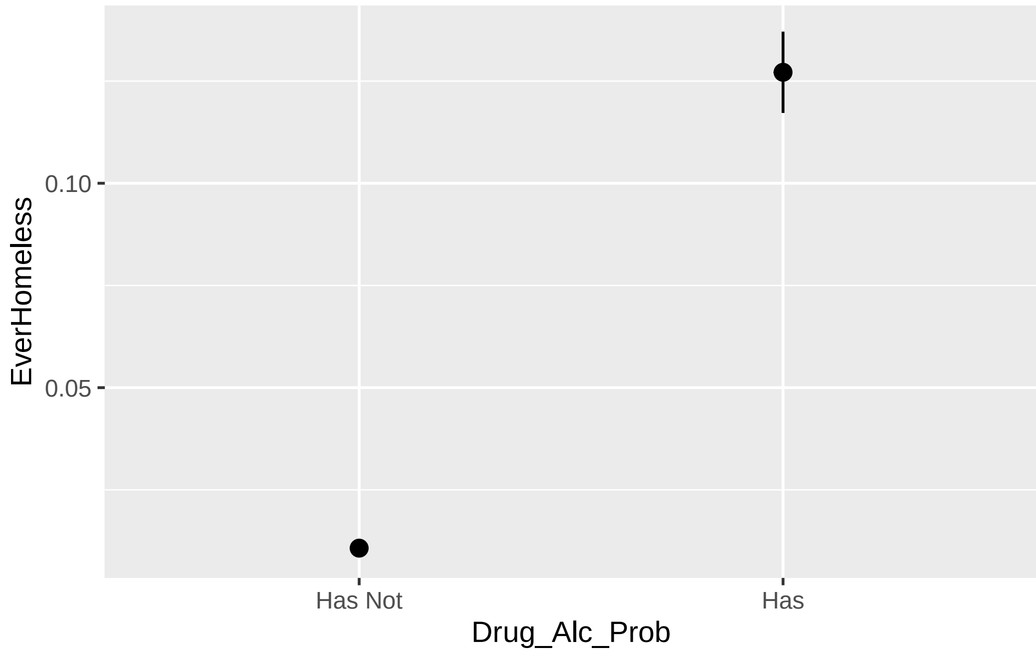
```
- Age            1   8709.4 8725.4
- Race           5   8948.1 8956.1
- Drug_Alc_Prob  1   9845.3 9861.3


Call:  glm(formula = EverHomeless ~ Drug_Alc_Prob + Age + Sex + Race,
    family = binomial(link = "logit"), data = no_na)

Coefficients:
          (Intercept)       Drug_Alc_ProbHas                    Age
            -5.273378               2.481524               0.057365
            SexFemale              RaceBlack    RaceNative American
            -0.162963               1.553791               1.348348
            RaceAsian   RacePacific Islander  RaceTwo or More Races
            -0.006418               1.391054               0.731884

Degrees of Freedom: 52730 Total (i.e. Null);  52722 Residual
Null Deviance:      10440
Residual Deviance: 8632     AIC: 8650
```

This appears to already be the most parsimonious version of the model as shown by both tests.

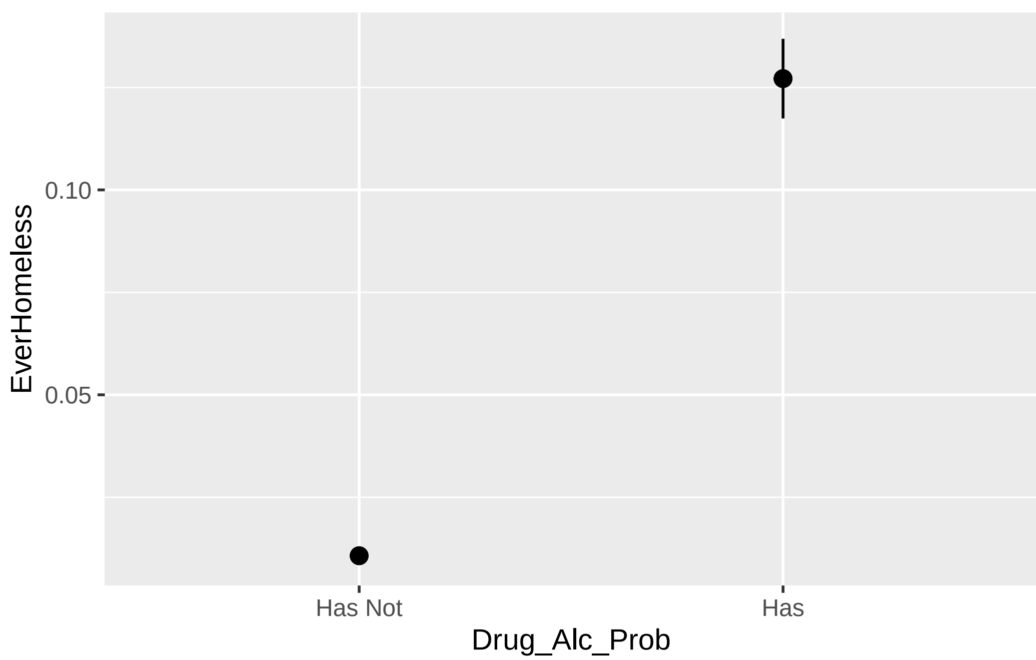
Plotting Our Data

Plotting our data to see if the effects of living with someone with a drug/alcohol problem on ever being homeless change significantly when using the unadjusted vs. adjusted models.

```
plot_predictions(model1, by = "Drug_Alc_Prob")
```




```
plot_predictions(model2, by = "Drug_Alc_Prob")
```



These plots are nearly identical, showing that this relationship holds even when adjusting for other covariates such as age, sex, and race.

Conclusions

There appears to be a strong association between a child having lived with someone with a drug/alcohol problem and having experienced homelessness. This association has a clinically significant effect size; however, the covariates do not affect the model sufficiently to be considered majorly impactful.

Future models should attempt to adjust for more covariates and include these, as they are included in the most parsimonious model. Additionally, future data collection looking at solely this relationship would likely help increase the power of these models, therefore allowing for more conclusive findings.